1.61 Large Data Set: OCR MEI (World Bank)

Worked Solutions

1 a)

USA

1 b)

North Africa, South Africa, Central America, North America, South America, Asia. Asia/Europe, Caribbean, Europe. Oceania

1 c)

It is based on statistics from population censuses, vital statistics registration systems, or sample surveys pertaining to the recent past and on assumptions about future trends.

1 d)

Square kilometres

2)

It is the average number of births (or deaths) per year per 1000 population at midyear.

3 a)

GDP is a measure of all the goods and services produced within a nation in a year.

3 b)

It is the purchasing power parity basis divided by population in 1 July for that year. Purchasing power measures the GDP in dollars based on comparing prices between the country and the US rather than using the official exchange rate between currencies.

4)

the number of medical doctors (physicians), including generalist and specialist medical practitioners, per 1000 of the population. Medical doctors are defined as doctors that study, diagnose, treat, and prevent illness, disease, injury, and other physical and mental impairments in humans through the application of modern medicine. They also plan, supervise, and evaluate care and treatment plans by other health care providers.

5 a)

physicians, nurses, and midwives only

5 b)

The World Health Organization estimates that fewer than 2.3 health workers per 1,000 would be insufficient to achieve coverage of primary healthcare needs.

5 c)

Albania, Bosnia and Herzegovina or Cyprus

5 d)

North Africa

6)

Current Health Expenditure (CHE) describes the share of spending on health in each country relative to the size of its economy. It includes expenditures corresponding to the final consumption of health care goods and services and excludes investment, exports, and intermediate consumption. CHE shows the importance of the health sector in the economy and indicates the priority given to health in monetary terms

7 a)

Sometimes the number in 'mobile phone subscribers per 100 population' is greater than 100 which shows that there are countries in which people have more than one phone per person on average.

7 b)

Oceania

7 c)

There are 236 countries in total so he approximately wants a $\frac{1}{5}$th of the data set. He should randomly sample 1 per 5 countries in each of the regions.

Eg. Select randomly the number below from each of the regions

North Africa: 1, South Africa: 11, Central America: 1, North America: 1, South America: 2, Asia: 10, Asia/Europe: 0, Caribbean: 5, Europe: 10, Oceania: 5

8 a)

No

8 b)

Yes – Lesotho

9 a)

Kosovo and Monaco are the outliers in the Europe boxplot Japan is the outlier in the Asia boxplot

9 b)

The median age is not available for the Holy See (Vatican City) and this is identifiable with a #N/A

9 c)

The median age of a person in Europe is higher on average and more consistent.

9 d)

Russia is categorised as Asia/Europe

10 a)

Mean is 5.88 % sample standard deviation is 1.81 % (population standard deviation is 1.74 %)

10 b)

$5.88 - 2 \times 1.81 = 2.26$ % $5.88 + 2 \times 1.81 = 9.5$ % (if using population: $5.88 - 2 \times 1.74 = 2.4$ % $5.88 + 2 \times 1.74 = 9.36$ %) Cuba with CHE 11.3 % is an outlier.

10 c)

Countries on average spend more per capita in Oceania then they do in the Caribbean, but there is less consistency in this in Oceania.

10 d)

CHE is a measure of what % of the countries expenditure is Both regions contain lots of smaller islands so may face similar challenges in providing healthcare.

10 e)

Oceania includes Australia and New Zealand which are commonly known as first world countries due to being highly industrialized, possessing a low poverty rate, and/or high accessibility to modern resources and infrastructure. Countries such as these tend to have higher CHE. If you were wanting to only compare small islands, you may want to choose to not include higher population countries.

11 a)

Botswana: $2.01 \times 3.5 + 50.1 = 57.135$ Chilie: $3.90 \times 3.5 + 55.5 = 69.15$

11 b)

Botswana: 50.1 is the life expectancy of someone in 1960 2.01 is the rate at which life expectancy increases per decade Chile: 55.5 is the life expectancy of someone in 1960 3.90 is the rate at which life expectancy increases per decade

11 c)

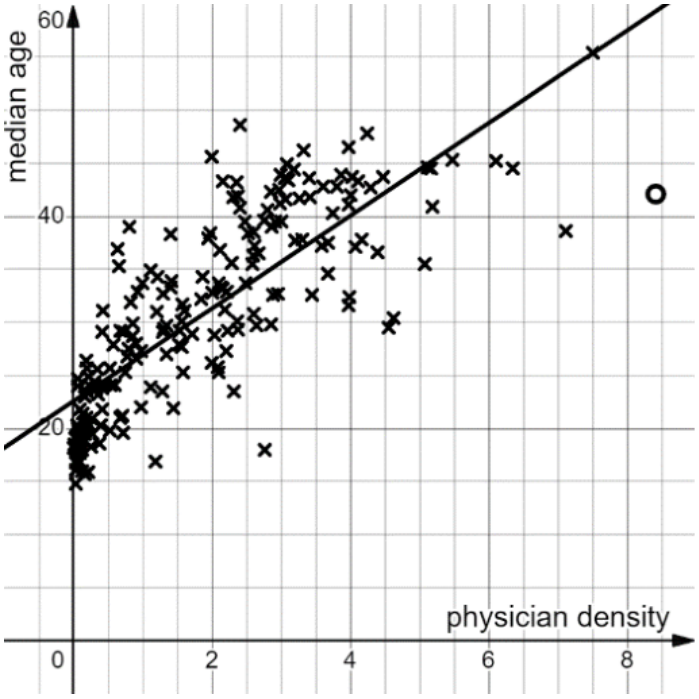
According to the models, Chile has a higher life expectancy in 1960 and has a higher rate of improvement in life expectancy than Botswana.

12 a)

The value of r indicates a strong positive linear relationship. However, median age will be limited by the age humans can reach, so it cannot be linear for higher values of physician density.

12 b)

The right most point, identified with a circle below:



12 c)

$y = 4.37 \times 7 + 22.6 = 53.19$ years

12 d)

Whilst it is interpolation (just) having a median age of 53.19 is high, which indicates a much older population. Therefore it is unlikely that physician density is the only factor they need to change to have a sustainable older median age.

13 a)

$\mathrm{P}(X > 6) = 0.7157$ (4 d.p.)

13 b)

$$\mathrm{P}(7.5 - 2 \times \sqrt{6.92} < X < 7.5 + 2 \times \sqrt{6.92}) = \mathrm{P}(2.2388 < X < 12.761)$$
$$= 0.9545 \text{ (4 dp)}$$

13 c)

$$\mathrm{P}(X > a) = 0.1 \Rightarrow \mathrm{P}(X < a) = 0.9$$
$$\therefore a = 10.9 \text{ (3 SF)}$$